



Analysis, state estimation, and control for the networked competitive multi-virus SIR model[☆]

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ABSTRACT

This paper proposes a novel discrete-time multi-virus susceptible–infected–recovered (SIR) model that captures the spread of competing epidemics over a population network. First, we provide sufficient conditions for the infection levels of all the viruses over the networked model to converge to zero in exponential time. Second, we propose an observation model that captures the summation of all the viruses' infection levels in each node, which represents the individuals who are infected by different viruses but share similar symptoms. Third, we present a sufficient, but not necessary, condition for the model to be strongly locally observable, and, assuming that the network has only infected or recovered individuals, a necessary and sufficient condition for the model to be strongly locally observable. We prove that the estimation error of our proposed estimator converges to zero asymptotically with the observer gain. Finally, we present a distributed feedback controller which guarantees that each virus dies out at an exponential rate. We then show via simulations that the estimation error of the Luenberger observer converges to zero before the viruses die out.

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1. Introduction

The history of human civilization has been a narrative of undergoing, battling, and outmatching various pandemics (Benedictow & Benedictow, 2004; Cheng & Leung, 2007; Johnson & Mueller, 2002). The devastation that epidemics leave in their wake is well-known; as examples, the 2009 swine flu pandemic, which was caused by the H1N1 influenza virus, resulted in 60.8 million cases worldwide (World Health Organization, 2022), and the more recent SARS-CoV-2 virus infected 642 million individuals globally (World Health Organization, 2020). Given the detrimental impacts that epidemics have on society, research on the spread of multiple diseases has attracted attention from several research communities. In fact, research activity has grown massively with the emergence of each epidemic. Various disease

spread models have been proposed and studied in the literature to capture epidemic processes based on the characteristics of individual pathogens; some of the most basic ones being the susceptible–infected–susceptible (SIS), susceptible–infected–recovered (SIR), and susceptible–infected–recovered–susceptible (SIRS) models (Mei, Mohagheghi, Zampieri, & Bullo, 2017). The SIR model is a simple yet classic epidemic model for diseases in which hosts recover with permanent or close to permanent immunity following infection. This model has been pivotal in providing a deeper understanding of a wide range of airborne diseases: Spanish Flu, SARS (Colizza, Barrat, Barthélemy, & Vespignani, 2007), MERS (Chang, 2017), Influenza (Osthus, Hickmann, Caragea, Higdon, & Del Valle, 2017), H1N1 (Coburn, Wagner, & Blower, 2009), Ebola (Berge, Lubuma, Moremedi, Morris, & Kondera-Shava, 2017), and COVID-19 (Calafiore, Novara, & Possieri, 2020; Chen, Lu, Chang, & Liu, 2020), to cite a few. In particular, the 2009 swine flu pandemic, which was caused by the H1N1 influenza virus, resulted in 60.8 million cases worldwide (World Health Organization, 2022), and the more recent SARS-CoV-2 virus, infected 642 million people globally (World Health Organization, 2020). The networked single-virus SIR model has been studied extensively, e.g., Ito (2022), She, Sundaram, and Paré (2022), Smith and Bullo (2022), Zhang, Leung, Butler, and Paré (2024).

Note that it is not unusual for multiple viruses/strains to simultaneously circulate in a community; for instance, different

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strains of Dengue fever (Pepin, Lambeth, & Hanley, 2008) and coronavirus (Poland, Vennema, Foley, & Pedersen, 1996) have been observed to be competing, based on real-world data. When there are multiple viruses circulating, there are two possibilities: (i) the viruses could either be cooperative, in which case, assuming there are two viruses simultaneously circulating, infection by one virus influences the likelihood of simultaneous infection by another virus (Gracy et al., 2025); or (ii) they could be competitive, in which case infection by one virus precludes the possibility of simultaneous infection by another virus (Gracy, Anderson, Ye, & Uribe, 2024; Liu, Paré, Nedić, Tang, Beck et al., 2019; Sahneh & Scoglio, 2014). A recent paper explores the setting of multi-competitive viruses where the population has partial or waning immunity (Zino, Ye, & Anderson, 2024). Inspired by the work on a single-population competing spread model in Nowak (1991), we investigate in this paper the spread of multi-competitive viruses over a population network. Note that the network model can efficiently capture the spread of epidemics over large populations as compared to single-population compartmental models (Van Mieghem, Omic, & Kooij, 2009). An extensive amount of effort has been expended on the study of multi-virus models (Paré, Liu, Beck, Nedić, & Başar, 2017; Paré, Vrabac, Sandberg, & Johansson, 2020; Prakash, Beutel, Rosenfeld, & Faloutsos, 2012; Sahneh & Scoglio, 2014; Santos, Moura, & Xavier, 2015), all of which focus on the competing SIS networked virus model, such as modeling the spread of competitive opinions spread over social networks (Sahneh & Scoglio, 2014), rival merchandise's sales in a market (Prakash et al., 2012), and competing farm subsidy programs from the US Department of Agriculture (Paré et al., 2020).

To the best of our knowledge, a networked multi-competitive SIR model that accounts for the presence of an arbitrary but finite number of viruses has not yet been studied in the literature. Additionally, the vast majority of the literature on epidemic models is in continuous time. However, employing a discrete-time model confers multiple advantages: (i) it potentially enables an easier comparison of experimental data with the predictions of a model, insofar as these predictions are given in discrete form, and (ii) the numerical exploration of discrete-time epidemic models is fairly straightforward, which implies that such models can be implemented with relative ease, even by those not mathematically oriented. The latter is significant in the context of public health (Brauer, Feng, & Castillo-Chavez, 2009). Thus, in this work, we propose a discrete-time multi-competitive networked SIR model. The multi-virus model captures the spread of multiple viruses/variants over a population, such as different variants of SARS-CoV-2 virus. Indeed, it is quite possible for the phenomenon of co-infection (and immune suppression) to manifest when there is a simultaneous spread of multiple viruses. However, to do so would require deeper analysis, since the dimension of the state space grows in a nonlinear fashion as the number of viruses increases, and is, therefore, beyond the scope of the present paper. We note that the model we introduce in this paper is related to the models in Allen and van den Driessche (2008), Van den Driessche (2008); see Remark 1 and Remark 2, respectively, for more details.

Beyond the modeling and analysis of epidemic models, monitoring of epidemics and estimation of infection levels in the population have been some of the fundamental quests in the research on competing contagions. Given that the SARS-CoV-2 pandemic has provided us with an enormous amount of data, the question of how to accurately infer the infection levels with respect to various strains of a virus (Delta, Omicron, etc.) and/or with respect to different viruses (influenza, SARS-CoV-2) has been pursued by the research community (Meyerowitz-Katz & Merone, 2020; Russell et al., 2020). Note that certain infectious diseases, in

particular Delta and Omicron variants of the SARS-CoV-2 virus, in spite of being competing, exhibit similar symptoms (Antonelli, Pujol, Spector, Ourselin, & Steves, 2022; Rader et al., 2022). Consequently, it is quite likely that an individual suffering from, for instance, the Omicron variant could be diagnosed as being infected by the Delta variant, and vice-versa. Thus, certain competing SIR viruses can affect the measurement of the cases corresponding to each virus and, therefore, pose difficulties for the estimation of the states (i.e., infection levels) with respect to each virus. When testing facilities are limited, as was witnessed at the beginning of the SARS-CoV-2 pandemic and at various peaks of different waves (Perks & Schneck, 2021), it is even more difficult to infer which (if any) virus is it that said individual is infected by (Belongia & Osterholm, 2020; Ma et al., 2021). Motivated by these real-world scenarios, in this paper, we consider the following question: assuming that there are multiple competing viruses spreading over a network, given measurements of individuals who exhibit symptoms corresponding to at least one (but possibly more) of the viruses, can we accurately estimate the infection level with respect to each virus? State estimation of nonlinear systems having a structure similar to SIR epidemic models has been studied in (Arcak & Kokotović, 2001; Bara, Zemouche, & Boutayeb, 2005; Niazi & Johansson, 2022). However, the scenario where the observation is an accumulation of the system states and has dimension lower than that of the system states, has not been investigated in the literature.

Assuming that one can identify a sufficient condition for estimation of the infection level with respect to each virus, we pose the following question: *how* can state estimation be carried out? The paper (Mitra & Sundaram, 2018) proposed a distributed observer for discrete-time LTI networked systems, where the observation model collects measurements of the system states from a node and its neighbors. That work, however, focuses on LTI systems, and hence the results there are not applicable to SIR models, which are nonlinear; the paper also did not consider the scenario when the dimension of the observation vector is lower than that of the state space. Nevertheless, inspired by the approach for observer design in (Mitra & Sundaram, 2018), we design here a distributed estimator which considers the measurements of the local node and the inferred systems states of the neighboring nodes in the network for estimation. Different from (Mitra & Sundaram, 2018), however, we assume that each node (corresponding to a subpopulation in our case) obtains completely accurate information regarding the values that the states of its neighbors take and that there is no time delay.

Note that the discussion so far has focused on the stability analysis, observability, and observer design for the discrete-time multi-competitive SIR model. Yet another challenging problem that policymakers are confronted with during the course of a pandemic is the design of effective eradication/mitigation strategies. As we show later in the paper, all the viruses in the competing SIR networked model die out eventually, regardless of the system parameters. However, the speed with which viruses die out is crucial; a faster speed guarantees that fewer individuals become infected by the virus over the course of the outbreak. That is, conditions guaranteeing exponential decay of viruses are preferable (Zhang et al., 2024). Moreover, mitigation strategies that account for the current infection levels and can be implemented at the node level are clearly more advantageous, thus motivating the need for distributed feedback control schemes. To the best of our knowledge, the design of distributed feedback controllers for competing SIR networked epidemic models has remained an open problem; the present paper aims to address this gap.

Through this paper, we make the following contributions:

1. We propose a discrete-time competitive multi-virus networked SIR model; see Eq. (2).

2. We provide sufficient conditions for the infection level of each virus to converge to zero in exponential time; see [Theorem 1](#) and [Proposition 1](#).
3. We specify a sufficient condition that guarantees that the system with two viruses is strongly locally observable at any arbitrary system state; see [Theorem 2](#). We show via simulations that the condition is actually not necessary; see [Remark 7](#).
4. We provide a necessary and sufficient condition that guarantees that the system is strongly locally observable at a certain system state; see [Theorem 3](#).
5. We propose a distributed state estimator and study how to design the observer gain such that the system state estimation errors converge to zero asymptotically; see [Theorem 4](#).
6. We propose a distributed mitigation algorithm that ensures that the infection level of each virus converges to zero in exponential time; see [Theorem 5](#).

This paper substantially expands upon (Zhang, Gracy, Başar, & Paré, 2022), where some of the results in this paper were presented. Specifically, [Theorems 2–5](#), and additional simulations, which corroborate the theoretical results of the paper and show how to choose the feasible observer gain such that the estimated system states are well-defined were not included in [Zhang et al. \(2022\)](#).

1.1. Paper outline

The paper is organized as follows: We conclude the present section by collecting all the notations that we would need in the sequel. Section 2 proposes a novel discrete-time competitive multi-virus networked SIR model, formally introduces the questions that the present paper seeks to address and provides preliminary results, which aid in the development of the main findings of this paper. Section 3 provides two sufficient conditions for eradication of a virus in exponential time. Section 4 examines the observation model, and is split into two: Section 4.1 specifies a sufficient condition for the system to be strongly locally observable at an arbitrary system state, and a necessary and sufficient for the system to be strongly locally observable at a specific system state (namely, where every node is either in the infected or in the recovered compartment). Section 4.2 proposes a distributed Luenberger estimator, which estimates the states of the aforementioned model, and determines a condition which ensures that the estimation error of the proposed observer converges to zero asymptotically. Section 5 proposes a distributed feedback controller that guarantees that each virus dies out at an exponential rate. In Section 6, we utilize simulations to illustrate the results from Sections Section 3, 4, and 5. Finally, Section 7 summarizes the main findings of the paper, and identifies some directions for future work. All proofs, except that of [Theorem 1](#), are provided in the Appendix.

1.2. Notation

We denote the set of real numbers and the set of non-negative integers by $\mathbb{R}$ and $\mathbb{Z}_{\geq 0}$, respectively. For any positive integer n , we have $[n] := \{1, 2, \dots, n\}$. The spectral radius and the maximum singular value of a matrix $A \in \mathbb{R}^{n \times n}$ are denoted by $\rho(A)$ and $\sigma_1(A)$, respectively. A diagonal matrix is denoted by $\text{diag}(\cdot)$. The transpose of a vector $x \in \mathbb{R}^n$ is $x^\top$. The Euclidean norm is denoted by $\|\cdot\|$ and the one norm is denoted by $|\cdot|$. We use I to denote the identity matrix. We use $\mathbf{0}$ and $\mathbf{1}$ to denote the vectors whose entries are all 0 and 1, respectively, where the dimensions of the vectors are determined by context. Given a matrix A , $A > 0$ indicates that A is positive definite, whereas $A < 0$ indicates that A is negative definite, and A^{-1} denotes the inverse of matrix A .

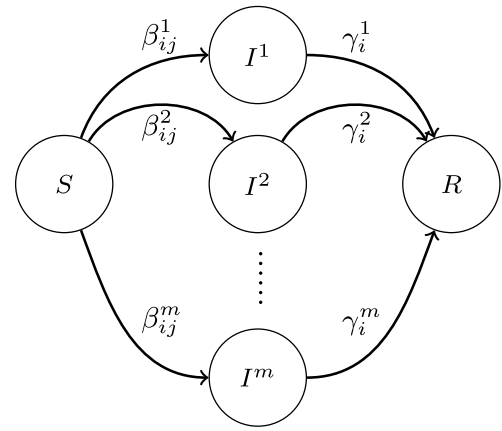


Fig. 1. The competing SIR networked model. Each node in the network can only be in one of $(m + 2)$ states: S, I^k, R , where $k \in [m]$.

2. Model and problem formulation

In this section, we present our system model, formally state the questions that the paper addresses, and recall some preliminary results that are used in the sequel.

2.1. System model

Consider a network of $n \geq 2$ nodes in which m ($m \geq 2$), viruses compete to infect the nodes. In context, a node here represents a well-mixed¹ population (hereafter, referred to as a subpopulation) of individuals with a large and constant size. Each individual is in one of $m + 2$ mutually exclusive compartments: susceptible, infected by virus k , $k \in [m]$, or recovered. We say that a node is healthy if all individuals within it are healthy; otherwise, we say that it is infected. An individual in the susceptible compartment transitions to the “infected by virus k ” (for $k \in [m]$) compartment depending on its infection rates with respect to virus k , namely β_{ij}^k (described in more detail below). An individual in the “infected by virus k ” compartment transitions to the recovered compartment depending on its recovery rate with respect to virus k , namely γ_i^k .

We use an m -layer, directed graph G to capture the spread of the m competing viruses. The vertices of G correspond to the population nodes. The infection graph that represents the pathways through which virus k , for each $k \in [m]$, spreads in the population is denoted by the k^{th} layer. In particular, there exists a directed edge from node j to node i in layer k if, assuming an individual in population j is infected by virus k , then said individual can infect at least one healthy individual in node i . The edge set corresponding to the k^{th} layer of G is denoted by E^k , while B^k denotes the weighted adjacency matrix (where $\beta_{ij}^k \geq 0$). Note that $(i, j) \in E^k$ if, and only if, $a_{ji}^k \neq 0$. Further, we allow the graphs across which each virus spreads to possibly be different since the transmission of one virus could occur through different means than another and/or an individual could be less susceptible to one virus over another.

We denote by s_i and r_i , respectively, the susceptible and recovered proportions of subpopulation i . We use $x_i^k[t]$, where $k \in [m]$, to denote the fraction of individuals at node i infected by virus k at time instant t . A pictorial depiction of this model is given in

¹ Well-mixed means that the probability of any two individuals, say i, j , interacting with each other is the same for all pairs (i, j) .

Fig. 1. In continuous time, the dynamics of the i th node can be written as follows:

$$\dot{s}_i = -(1 - x_i^1 - \dots - x_i^m - r_i) \sum_{k=1}^m \sum_{j=1}^n \beta_{ij}^k x_j^k, \quad (1a)$$

$$\dot{x}_i^k = (1 - x_i^1 - \dots - x_i^m - r_i) \sum_{j=1}^n \beta_{ij}^k x_j^k - \gamma_i^k x_i^k, \quad (1b)$$

$$\dot{r}_i = \sum_{k=1}^m \gamma_i^k x_i^k, \quad \forall i \in [n]. \quad (1c)$$

Note that while an epidemic process evolves in continuous time, the data regarding the evolution of an epidemic are compiled on a daily basis (Snow, 1855; World Health Organization, 2020) or on a weekly basis (World Health Organization, 2022). Such a sampling of the system behavior motivates the use of a discrete-time multi-competitive networked SIR model. By using the Euler method (Atkinson, 2008), we derive the discrete-time dynamics of the SIR networked epidemic model at node i as:

$$s_i[t+1] = s_i[t] - h \left(s_i[t] \sum_{k=1}^m \sum_{j=1}^n \beta_{ij}^k x_j^k[t] \right), \quad (2a)$$

$$x_i^k[t+1] = x_i^k[t] + h \left(s_i[t] \sum_{j=1}^n \beta_{ij}^k x_j^k[t] - \gamma_i^k x_i^k[t] \right), \quad (2b)$$

$$r_i[t+1] = r_i[t] + h \sum_{k=1}^m \gamma_i^k x_i^k[t], \quad (2c)$$

where $h > 0$ is the sampling parameter, t is the time index, and $k \in [m]$ indicates the k^{th} virus. Notice that $s_i[t] + x_i^1[t] + \dots + x_i^m[t] + r_i[t] = 1$, capturing the fact that in the competing virus scenario, all the viruses are mutually exclusive. We now rewrite (2) in compact form as:

$$s[t+1] = hS[t] \sum_{k=1}^m B^k x^k[t] \quad (3a)$$

$$x^k[t+1] = x^k[t] + h[S[t]B^k - \Gamma^k]x^k[t], \quad (3b)$$

where $S[t] = \text{diag}(s_i[t])$ and $\Gamma^k = \text{diag}(\gamma_i^k)$. Note that $r[t]$ can be omitted since the states add up to one.

The models in (1) and (2) have close connections with mathematical models introduced for studying disease spread by taking into account spatial variation, as we discuss in the following remarks.

Remark 1. By setting $m = 1$ (i.e., no competition), the model in (1) coincides with (Van den Driessche, 2008, model (7.2)).

Remark 2. Setting $m = 1$ and $n = 2$, the model in (2) yields an approximation of (Allen & van den Driessche, 2008, model (20)).

We now introduce the following assumptions to ensure that the model in (2) is well-defined.

Assumption 1. For all $i \in [n]$ and $k \in [m]$, we have $s_i[0], x_i^k[0], r_i[0] \in [0, 1]$ and $s_i[0] + \sum_{k=1}^m x_i^k[0] + r_i[0] = 1$.

Since the states are proportions of the subpopulations, they must be between zero and one, and add up to one.

Assumption 2. For all $i, j \in [n]$, and $k \in [m]$, we have $\beta_{ij}^k \geq 0, \gamma_i^k > 0, h > 0$.

Since β_{ij}^k, γ_i^k are rates, they must be nonnegative. To ensure that we have an SIR model, and not SI, the recovery rates must be positive. In addition, the sampling rate h must be positive; otherwise, the states will never evolve.

Assumption 3. For all $i \in [n]$, we have $h \sum_{k=1}^m \sum_{j=1}^n \beta_{ij}^k \leq 1$ and $h \sum_{k=1}^m \gamma_i^k \leq 1$.

This assumption ensures that the sampling parameter h is small enough to guarantee that the states do not go below zero or above one.

Assumptions 1 and 2 can be interpreted as the initial proportion of susceptible, infected, and recovered individuals, all lying in the interval $[0, 1]$, and that the healing rates are always positive, which are standard in the literature (Brauer, Castillo-Chavez, & Feng, 2019; Mei et al., 2017; She, Leung, Sundaram, & Paré, 2021). **Assumption 3** ensures that the sampling rate is frequent enough for the states of the model to remain well-defined.

Motivated by the fact that different infectious diseases can demonstrate similar symptoms over the hosts (Belongia & Osterholm, 2020; Ma et al., 2021), we build an observation model which produces the output as the aggregated proportion of individuals who show flu-like symptoms from infection of all viruses. The observation model is written as (where we repeat (2b) for convenience):

$$x_i^k[t+1] = x_i^k[t] + h \left\{ s_i[t] \sum_{j=1}^n \beta_{ij}^k x_j^k[t] - \gamma_i^k x_i^k[t] \right\}, \quad (4a)$$

$$y_i[t] = \sum_{k=1}^m c_i^k x_i^k[t], \quad (4b)$$

where c_i^k is the measurement coefficient.

Assumption 4. The coefficient $c_i^k \in (0, 1]$ for all $i \in [n], k \in [m]$.

Remark 3. The coefficient c_i^k from Eq. (4b) can capture the probability of some individual in subpopulation i showing symptoms associated with virus k . Therefore, $1 - c_i^k$ captures the probability of some individuals infected by virus k in subpopulation i being asymptomatic. The probability of exhibiting symptoms corresponding to different viruses has been studied in, among others, (Panovska-Griffiths et al., 2020; Teunis et al., 2008). The coefficient c_i^k can also represent how subpopulation i defines and measures the cases based on the symptoms of each virus k . For example, the symptoms of the SARS-CoV-2 virus can include but are not limited to fever, muscle aches, cough, runny nose, headaches, and fatigue.

Remark 4. Given a subpopulation i , Eq. (4b) represents the total proportion of symptomatic patients in subpopulation i . The left-hand side of Eq. (4b), i.e., $y_i[t]$, is a useful indicator for decision-makers to formulate policies regarding the adequacy of local healthcare facilities and the availability of virus-combating resources.

We present the following results for system (2).

Lemma 1. Under Assumptions 1–3, for all $i \in [n]$ and $t \in \mathbb{Z}_{\geq 0}$,

1. $s_i[t], x_i^k[t], r_i[t] \in [0, 1]$, for all $k \in [m]$, and $s_i[t] + \sum_{k=1}^m x_i^k[t] + r_i[t] = 1$, and
2. $s_i[t+1] \leq s_i[t]$.

2.2. Problem formulation

With the system model set up as in Section 2.1, we now introduce the problems investigated in this paper.

- (i) For the system with dynamics given in (3), provide a sufficient condition which ensures that, for some $k \in [m]$, $x^k[t]$ converges to the eradicated state, namely $x^k = \mathbf{0}$, in exponential time. Provide an upper bound on the rate of convergence for the sequence $x^k[t], k \in [m]$ (converging to $\mathbf{0}$ exponentially).

- (ii) Provided that the infection rate matrix B^k , healing rate matrix Γ^k , the observation $y_i[t]$, true susceptible states $s_i[t]$, and the measurement coefficient c_i^k , are known, for all $i \in [n]$, $k \in [m]$, under what conditions are the infection levels of each virus $x_i^k[t]$, for all $i \in [n]$, $k \in [m]$ strongly locally observable, at $s_i[t] = 0$, $\forall i \in [n]$?
- (iii) Given the system parameters β_{ij}^k, γ_i^k for all $i, j \in [n]$, $k \in [m]$, and the local aggregated observation, $y_i[t]$ for all $i \in [n]$, construct a distributed Luenberger observer which delivers the system states $\hat{s}_i[t]$, $\hat{x}_i^k[t]$, $\hat{r}_i[t]$ for all $i \in [n]$, $k \in [m]$.
- (iv) Given the system parameters β_{ij}^k, γ_i^k for all $i, j \in [n]$, $k \in [m]$, how do we find the gain of the distributed observer such that the estimation error $x^k[t] - \hat{x}^k[t]$ converges to zero asymptotically for all $k \in [m]$?
- (v) Can a distributed feedback controller be designed such that all viruses are eradicated at an exponential rate?

2.3. Preliminaries

In this subsection, we recall certain preliminary results from nonlinear systems theory that will help in the development of the main results of the paper.

Consider a system described as follows:

$$x[t+1] = f(t, x[t]), \quad (5a)$$

$$y[t] = g(x[t]), \quad (5b)$$

where $y[t]$ is the output of the system.

Definition 1. An equilibrium point of (5a) is globally exponentially stable (GES) if there exist positive constants α and ω , with $0 \leq \omega < 1$, such that

$$\|x[t]\| \leq \alpha \|x[t_0]\| \omega^{(t-t_0)}, \forall t \geq t_0 \geq 0, \forall x[t_0] \in \mathbb{R}^n. \quad (6)$$

We recall a sufficient condition for GES of an equilibrium of (5a) from Vidyasagar (2002).

Lemma 2. (Vidyasagar, 2002, Theorem 28) Suppose that there exist a function $V : \mathbb{Z}_+ \times \mathbb{R}^n \rightarrow \mathbb{R}$, and constants $a, b, c > 0$ and $p > 1$ such that $a\|x\|^p \leq V(x) \leq b\|x\|^p$, $\Delta V(x) := V(x[t+1]) - V(x[t]) \leq -c\|x\|^p$, $\forall t \in \mathbb{Z}_{\geq 0}$. Then, $\forall x(t_0) \in \mathbb{R}^n$, $x = \mathbf{0}$ is the globally exponential stable equilibrium of (5a).

Note that Lemma 2 guarantees convergence to an equilibrium of (5a); however, it does not say anything about the rate at which convergence to the equilibrium occurs. The following lemma addresses the same.

Lemma 3. (Rugh, 1996, Theorem 23.3) Under the conditions of Lemma 2, convergence to the origin has an exponential rate of at least $\sqrt{1-(c/b)} \in [0, 1)$, where b and c are as defined in Lemma 2.

We need the following lemma to establish global exponential convergence to the set of healthy states.

Lemma 4. (Rantzer, 2011, Proposition 2) Suppose that M is a nonnegative matrix which satisfies $\rho(M) < 1$. Then, there exists a diagonal matrix $P > 0$ such that $M^\top P M - P < 0$.

The preliminary material has so far focused on system (5a). We now turn our attention towards the notion of strong local observability, by taking advantage of the set of measurements available via (5b). As a first step in this direction, we recall the definition of strong local observability of a nonlinear system.

Definition 2. The system in Eq. (4) is strongly locally observable at $s[t]$ if we are able to recover $x_i^k[t]$ for all $i \in [n]$, $k \in [m]$ through the sequence of outputs in the duration of $[t, t+m-1]$.

With Definition 2 in place, we recall a necessary and sufficient condition for strong local observability of a nonlinear system.

Lemma 5. (Nijmeijer, 1982; Sontag, 1979) The system in (5) is strongly locally observable at $x[t]$ if and only if the map $x[t] \rightarrow (g(x[t]), g(f^1(x[t])), \dots, g(f^{n-1}(x[t])))$ is injective, where n is the dimension of $x[t]$.

3. Healthy state analysis

In this section, we identify multiple sufficient conditions which guarantee that each virus k is eradicated exponentially fast, and provide the associated rates of convergence for each virus. Similar to the standard single-virus SIR networked model, the multi-competitive SIR networked model also converges to the set of healthy states regardless of (i) the values the system parameters take, and (ii) initial conditions. However, it is important to study the exponential convergence since it guarantees that the viruses die out at a faster rate and, as a consequence, fewer individuals become infected over the course of the outbreak (Zhang et al., 2024).

Define

$$M^k := I - h\Gamma^k + hB^k, \quad (7)$$

$$\tilde{M}^k[t] := I + h\{S[t]B^k - \Gamma^k\}, \quad (8)$$

and note that $\tilde{M}^k$ is the state matrix of system (3b):

$$\tilde{M}^k[t] = M^k - h(I - S[t])B^k. \quad (9)$$

We first present a sufficient condition, in terms of M^k , for the infection level with respect to each of the viruses to converge to zero exponentially fast.

Theorem 1. Consider system (2) under Assumptions 1–3. If $\rho(M^k) < 1$, then virus k gets eradicated in exponential time.

Proof. Consider an arbitrary virus $k \in [m]$. By Assumptions 2 and 3, M^k , defined in (7), is nonnegative for all $k \in [m]$. By assumption, $\rho(M^k) < 1$. Therefore, according to Lemma 4, for each $k \in [m]$, there exists a positive-definite diagonal matrix P^k such that $(M^k)^\top P^k M^k - P^k$ is negative definite.

Consider the candidate Lyapunov function: $V_1^k(t, x^k) = (x^k)^\top P^k x^k$. Since P^k is a positive-definite diagonal matrix, $(x^k)^\top P^k x^k > 0$, for all $x^k \neq \mathbf{0}$. Therefore, $V_1^k(t, x^k) > 0$, for all $k \in [m]$, $t \in \mathbb{Z}_{\geq 0}$, $x^k \neq \mathbf{0}$. Since P^k is positive definite,

$$\lambda_{\min}(P^k)I \leq P^k \leq \lambda_{\max}(P^k)I, \quad (10)$$

which implies that

$$\mu_1^k \|x^k\|^2 \leq V_1^k(t, x^k) \leq \mu_2^k \|x^k\|^2, \quad (11)$$

where $\mu_1^k = \lambda_{\min}(P^k)$ and $\mu_2^k = \lambda_{\max}(P^k)$, with $\mu_1^k, \mu_2^k > 0$.

Now we turn to computing $\Delta V_1^k(t, x^k)$. For $x^k \neq \mathbf{0}$ and for all $k \in [m]$, using (7) and (3b)–(8), we have

$$\begin{aligned} \Delta V_1^k(t, x^k) &= (x^k)^\top \tilde{M}^k[t]^\top P^k \tilde{M}^k[t] x^k - (x^k)^\top P^k x^k \\ &= (x^k)^\top [(M^k)^\top P^k M^k - P^k] x^k \\ &\quad - 2h(x^k)^\top (B^k)^\top (I - S[t]) P^k M^k x^k \\ &\quad + h^2 (x^k)^\top (B^k)^\top (I - S[t]) P^k (I - S[t]) B^k x^k. \end{aligned} \quad (12)$$

Note that the second and third terms of (12) can be reorganized as

$$(x^k)^\top [-2h(B^k)^\top (I - S[t]) P^k M^k$$

$$\begin{aligned}
& + h^2(B^k)^\top(I - S[t])P^k(I - S[t])B^k x^k \\
& = (x^k)^\top \{h(B^k)^\top(I - S[t])P^k \\
& \quad [-2M^k + h(I - S[t])B^k]\} x^k \\
& = (x^k)^\top \{h(B^k)^\top(I - S[t])P^k \\
& \quad [-2(I - h\Gamma^k) - h(I + S[t])B^k]\} x^k \leq 0,
\end{aligned} \tag{13}$$

where the last equality follows from (7), and the inequality follows from [Assumptions 2–3](#) and [Lemma 1](#). Thus, by plugging (13) into (12), we obtain

$$\Delta V_1^k(t, x^k) \leq (x^k)^\top [(M^k)^\top P^k M^k - P^k] x^k. \tag{14}$$

Since $[(M^k)^\top P^k M^k - P^k]$ is negative definite, we have, from Eq. (14),

$$\Delta V_1^k(t, x^k) \leq -\mu_3^k \|x^k\|^2, \tag{15}$$

where $\mu_3^k = \lambda_{\min}[P^k - (M^k)^\top P^k M^k]$, with $\mu_3^k > 0$.

Therefore, from (11) and (15), $V_1^k(t, x^k)$ is a Lyapunov function with exponential decay, and, hence, from [Lemma 2](#), $x^k[t]$ converges to zero at an exponential rate. ■

The proof of [Theorem 1](#) was inspired by the proof of ([Gracy, Paré, Sandberg, & Johansson, 2020](#), Theorem 1).

Note that $\rho(M^k)$ is the basic reproduction number of virus k . It is the average number of infections produced by an infected individual in a population where everyone is susceptible.

Applying [Lemma 3](#), the following is a direct consequence of [Theorem 1](#).

Corollary 1. Suppose that the conditions in [Theorem 1](#) are fulfilled, and that P^k is as defined in the proof of [Theorem 1](#). Then, the convergence of $x^k[t]$ has an exponential rate of at least $\sqrt{1 - \frac{\mu_3^k}{\mu_2^k}}$, where $\mu_2^k = \lambda_{\max}(P^k)$, $\mu_3^k = \lambda_{\min}[P^k - (M^k)^\top P^k M^k]$ for each $k \in [m]$.

In words, [Theorem 1](#) says that if the state matrix corresponding to (3b), linearized around $s_i = 1$ for $i \in [n]$, is Schur, then virus k , irrespective of its initial infection levels in the population, becomes extinct exponentially fast.

We now present another sufficient condition, which depends on $\sigma_1(\tilde{M}^k[t])$, the maximum singular value of $\tilde{M}^k[t]$, for the infection level with respect to each of the viruses to converge to zero at an exponential rate.

Proposition 1. Consider the system in (2) under [Assumptions 1–3](#). If $\sigma_1(\tilde{M}^k[t]) < 1$ for all $t \in \mathbb{Z}_{\geq 0}$, then the k th virus of the system in (2) converges to zero in exponential time, with at least an exponential rate of $\min_{t \in \mathbb{Z}_{\geq 0}} [1 - \sigma_1(\tilde{M}^k[t])]$.

4. State observation model

In this section, we analyze the observation model introduced in Eqs. (4a) and (4b). In particular, we focus on identifying conditions guaranteeing strong local observability of our proposed system model around $s_i[t] = 0$, and on estimating the system states with respect to each virus.

4.1. Strong local observability

As a first step, we construct the observability matrix for the system by writing Eq. (4b) as:

$$\mathbf{y}[t] = \mathbf{C}\mathbf{X}[t], \tag{16}$$

where $\mathbf{y}[t] = [y_1[t] \ y_2[t] \ \dots \ y_n[t]]^\top \in \mathbb{R}^n$; the measurement matrix $\mathbf{C} \in \mathbb{R}^{n \times mn}$ is defined as:

$$\mathbf{C} = [C^1 \ C^2 \ \dots \ C^m]$$

with $C^k = \text{diag}([c_1^k, c_2^k, \dots, c_n^k])$ for all $k \in [m]$; and

$$\mathbf{X}[t] = [x^1[t]^\top \ x^2[t]^\top \ \dots \ x^m[t]^\top]^\top.$$

Therefore, the measurement $\mathbf{y}[t]$ can be reorganized as:

$$\mathbf{y}[t] = C^1 x^1[t] + C^2 x^2[t] + \dots + C^m x^m[t]. \tag{17}$$

The measurements, corresponding to each time step, over a time horizon $[t, t + m - 1]$ can be concatenated in a vector as follows:

$$\begin{bmatrix} \mathbf{y}[t] \\ \mathbf{y}[t+1] \\ \mathbf{y}[t+2] \\ \vdots \\ \mathbf{y}[t+m-1] \end{bmatrix} = \underbrace{\begin{bmatrix} \mathcal{O}^1[t] & \mathcal{O}^2[t] & \dots & \mathcal{O}^m[t] \end{bmatrix}}_{\mathbb{O}[t]} \mathbf{X}[t], \tag{18}$$

where $\mathbb{O}[t] \in \mathbb{R}^{mn \times mn}$ is the observability matrix of the system in Eq. (4) and

$$\mathcal{O}^k[t] = \begin{bmatrix} C^k \\ C^k \tilde{M}^k[t] \\ C^k \tilde{M}^k[t] \tilde{M}^k[t+1] \\ \vdots \\ C^k \tilde{M}^m[t] \dots \tilde{M}^k[t+m-2] \end{bmatrix},$$

with $\mathcal{O}^k[t] \in \mathbb{R}^{mn \times n}$ for all $k \in [m]$.

We now present a sufficient condition for the system in (4) to be strongly locally observable for an arbitrary $s[t]$.

Theorem 2. Consider the system in Eq. (4) with $m = 2$ under [Assumptions 1–4](#). Suppose that $y_i[t]$ and $y_i[t+1]$, for all $i \in [n]$, are known. If

$$c_i^2 |\gamma_i^1 - \gamma_i^2 + s_i[t](\beta_{ii}^2 - \beta_{ii}^1)| > s_i[t] \sum_{j \neq i} |c_i^2 \beta_{ij}^2 - \frac{c_i^1 c_j^2}{c_j^1} \beta_{ij}^1|,$$

for all $i \in [n]$, then the system in Eq. (4) is strongly locally observable at $s[t]$.

As can be inferred from the proof and is shown via simulations in Section 6 (see [Remark 7](#)), the condition in [Theorem 2](#) is not necessary.

For the special case when $s[t] = 1$ (i.e., the beginning of an outbreak), [Theorem 2](#) implies the following result:

Corollary 2. Consider the system in Eq. (4) with $m = 2$ under [Assumptions 1–4](#). Suppose that $y_i[t]$ and $y_i[t+1]$, for all $i \in [n]$, are known. If

$$c_i^2 |\gamma_i^1 - \gamma_i^2 + (\beta_{ii}^2 - \beta_{ii}^1)| > \sum_{j \neq i} |c_i^2 \beta_{ij}^2 - \frac{c_i^1 c_j^2}{c_j^1} \beta_{ij}^1|,$$

for all $i \in [n]$, then the system in Eq. (4) is strongly locally observable at $s[t] = 1$.

We also have the following result for strong local observability at $s[t] = 0$ (i.e., the peak/end of a serious outbreak), which holds for any finite number of viruses m .

Theorem 3. Consider the system in Eq. (4) under [Assumptions 1–4](#). Suppose that $y_i[\tau]$, for all $i \in [n]$ and $\tau \in [t, t + m - 1]$ is known. The system in Eq. (4) is strongly locally observable at $s[t] = 0$ if and only if, for each $i \in [n]$, $\gamma_i^k \neq \gamma_i^{k_0}$ for all $k, k_0 \in [m]$ and $k \neq k_0$.

Note that the conditions in [Theorem 2](#) and in [Theorem 3](#) identify scenarios that admit the design of an observer for estimating the system states, but do not say how the states of the system can be estimated. Hence, in the sequel, we focus on the estimation of the system states.

4.2. State estimation

The dynamics of the estimated states are

$$\hat{x}_i^k[t+1] = \hat{x}_i^k[t] + h \left\{ \hat{s}_i[t] \sum_{j=1}^n \beta_{ij}^k \hat{x}_j^k[t] - \gamma_i^k \hat{x}_i^k[t] \right\}, \quad (19)$$

$$\hat{y}_i[t+1] = \sum_{k=1}^m c_i^k \hat{x}_i^k[t+1], \quad (20)$$

where

$$\hat{s}_i[t] = 1 - \sum_{k=1}^m \hat{x}_i^k[t] - \hat{r}_i[t] \quad (21)$$

and the recovered level is estimated by

$$\hat{r}_i[t] = h \sum_{q=0}^t \sum_{k=1}^m \gamma_i^k \hat{x}_i^k[q] \quad (22)$$

at each time step, recursively. Notice that in Eq. (19), in order to acquire the estimated infection level at node i , we need some knowledge of the infection levels from all the neighbors of node i , namely $\hat{x}_j^k[t]$ for all $\beta_{ij}^k \neq 0$. Hence, we have the following definition and assumption.

Definition 3. For node j , which is a neighbor node of node i , namely $\beta_{ij}^k > 0$, we define the estimated infection level at node j acquired by node i at time t as $\hat{x}_j^k[t - \tau_j] + \varepsilon_j$, where $\tau_j \in \mathbb{Z}_{\geq 0}$ is the time delay between nodes i and j , and $\varepsilon_j \in \mathbb{R}$ is the reporting error at node j .

Assumption 5. For the estimation algorithm, the nodes share their estimated infection levels with no time delay or error, namely $\tau_j = 0$, $\varepsilon_j = 0$ for all $j \in [n]$ in Definition 3.

Assumption 5 states that every node in the network is completely cooperative, and that it instantaneously shares their true infection levels with their neighbors. Hence, under Assumption 5, our proposed distributed Luenberger observer is:

$$\begin{aligned} \hat{x}_i^k[t+1] = & \hat{x}_i^k[t] + h \left\{ \hat{s}_i[t] \sum_{j=1}^n \beta_{ij}^k \hat{x}_j^k[t] - \gamma_i^k \hat{x}_i^k[t] \right\} \\ & + L_i^k (y_i[t] - \hat{y}_i[t]), \end{aligned} \quad (23)$$

where L_i^k is the observer gain which, given a node i , can be chosen for each $k \in [m]$. We can write $e_i^k[t] = x_i^k[t] - \hat{x}_i^k[t]$ as the error of the observer. Hence, the dynamics of the estimation error can be written as:

$$\begin{aligned} e_i^k[t+1] = & x_i^k[t+1] - \hat{x}_i^k[t+1] \\ = & (1 - h\gamma_i^k)e_i^k[t] + h \sum_{j=1}^n \beta_{ij}^k e_j^k[t] - L_i^k \sum_{k=1}^m c_i^k e_i^k[t] \\ & - h \left(\sum_{k=1}^m x_i^k[t] \sum_{j=1}^n \beta_{ij}^k x_j^k[t] - \sum_{k=1}^m \hat{x}_i^k[t] \sum_{j=1}^n \beta_{ij}^k \hat{x}_j^k[t] \right) \\ & - h \left(\sum_{q=0}^t \sum_{k=1}^m \gamma_i^k x_i^k[q] \sum_{j=1}^n \beta_{ij}^k x_j^k[q] \right. \\ & \left. - \sum_{q=0}^t \sum_{k=1}^m \gamma_i^k \hat{x}_i^k[q] \sum_{j=1}^n \beta_{ij}^k \hat{x}_j^k[q] \right). \end{aligned} \quad (24)$$

We then rewrite the error dynamics (24) as:

$$e^k[t+1] = (M^k - L^k C^k) e^k[t] + w^k[t] \quad (25)$$

where recall that

$$M^k = I - h\Gamma^k + hB^k$$

and

$$w^k[t] = -L^k \sum_{p \neq k}^m c^p e^p[t]$$

$$\begin{aligned} & + h \left\{ \left(\sum_{k=1}^m x^k[t] \right) B^k x^k[t] - \left(\sum_{k=1}^m \hat{x}^k[t] \right) B^k \hat{x}^k[t] \right\} \\ & + h \left\{ \sum_{q=0}^t \sum_{k=1}^m \Gamma^k x^k[q] B^k x^k[q] - \sum_{q=0}^t \sum_{k=1}^m \gamma_i^k \hat{x}_i^k[q] B^k \hat{x}^k[q] \right\} \end{aligned} \quad (26)$$

with $L^k = \text{diag}(L_i^k)$.

Inspired by Niazi and Johansson (2022), we aim to show that the estimation error of our Luenberger observer (23) converges to zero asymptotically. We make the following assumption.

Assumption 6. There exists a constant l^k for each virus $k \in [m]$ such that:

$$\|w^k[t]\| \leq l^k \|x^k[t] - \hat{x}^k[t]\| \quad (27)$$

for all $t \in \mathbb{Z}_{\geq 0}$.

Assumption 6 states that, for each virus k , we can upper bound the nonlinear terms and the terms that depend on the error of other viruses which appear in its error dynamics by a bounded scaling of its error $e^k[t]$. In practice, we are able to tune the observer gain L^k to satisfy the inequality (27) in Assumption 6. We explore the feasibility of Assumption 6 via simulations in Section 6. Note that, for each $k \in [m]$, (27) can be rewritten as:

$$l^k \geq \frac{\|w^k[t]\|}{\|x^k[t] - \hat{x}^k[t]\|}, \quad (28)$$

for all $t \in \mathbb{Z}_{\geq 0}$.

We now identify a sufficient condition, with respect to the observer gain, that guarantees that the estimation errors converge to zero asymptotically.

Theorem 4. Under Assumption 6, if there exist a symmetric matrix $Q^k > 0$ and $\tau^k \in (0, 1]$ such that the following inequalities are satisfied:

$$\begin{aligned} (M^k - L^k C^k)^\top [Q^k - (Q^k - \tau^k I)^{-1} Q^k] (M^k - L^k C^k) \\ - Q^k + \tau^k (l^k)^2 I < 0 \end{aligned} \quad (29)$$

and

$$Q^k - \tau^k I < 0. \quad (30)$$

where L^k is the observer gain for virus k , then the estimation error of the Luenberger observer (23) for virus k converges to zero asymptotically.

Remark 5. Note that we need the pair (M^k, C^k) to be detectable for inequalities (29) and (30) to hold (Bara et al., 2005; Niazi & Johansson, 2022). Under Assumption 4, the pair (M^k, C^k) is observable, and, thus, detectable.

Corollary 3. The inequalities (29) and (30) combined together in Theorem 4 are equivalent to the following LMI:

$$\begin{bmatrix} -Q^k + \tau^k (l^k)^2 I & (M^k)^\top Q^k - (C^k)^\top R^k & (M^k)^\top Q^k - (C^k)^\top R^k \\ [(M^k)^\top Q^k - (C^k)^\top R^k]^\top & Q^k - \tau^k I & \mathbf{0} \\ [(M^k)^\top Q^k - (C^k)^\top R^k]^\top & \mathbf{0} & -Q^k \end{bmatrix} < 0 \quad (31)$$

where $(R^k)^\top = Q^k L^k$.

By applying the Schur complement, LMI (31) can be shown to be equivalent to Eq. (45) and $-Q^k < 0$. Note that LMI (31) can be solved using the *cvx* solver in MATLAB for simulations. While the conditions in the LMIs (31) and (45) are not the most intuitive, they do provide a straightforward and effective method to estimate the system states, as illustrated via simulations in Section 6. Further, the observer for each virus k can be designed independently, as long as Assumption 6 is not violated.

5. Distributed feedback control

In this section, we present a distributed feedback mitigation strategy for ensuring that all viruses are eradicated. We establish that virus k can be eradicated in exponential time by boosting the healing rate associated with virus k . Applying such an eradication strategy for all m viruses, the system converges to a healthy state.

When battling against the spread of an epidemic, boosting the healing rate with respect to the virus, implemented by means of providing effective medication, medical supplies, and/or healthcare workers to each subpopulation, is a common approach (Gracy et al., 2020; Zhang et al., 2024). We formally state our distributed feedback control strategy as follows:

$$\tilde{\gamma}_i^k[t] = \gamma_i^k + u_i^k[t], \quad i \in [n], \quad (32)$$

where $u_i^k[t]$ is a state feedback controller, with

$$u_i^k[t] = s_i[t] \sum_{j=1}^n \beta_{ij}^k = \left(1 - \sum_{k=1}^m x_i^k[t] - r_i[t]\right) \sum_{j=1}^n \beta_{ij}^k. \quad (33)$$

Note that this proposed controller allocates resources to the most vulnerable populations (captured by the $\sum_{j=1}^n \beta_{ij}^k$ term) while they are still susceptible to infection (captured by the $s_i[t]$ term). While this control design assumes that there are no constraints on resources, it is an equitable way to allocate resources, in the sense that the more vulnerable susceptible subpopulations are provided with more public health/medical support. We have the following result.

Theorem 5. Consider the system in (2), under Assumptions 2 and 3, and assume further that $h\tilde{\gamma}_i^k[t] < 1$, $\forall i \in [n]$, $t \in \mathbb{Z}_{\geq 0}$. If

$$1 + h \left[\sum_{j \neq i}^n s_j[t] \beta_{ji}^k - \gamma_i^k \right] < 1,$$

$\forall i \in [n]$, $t \in \mathbb{Z}_{\geq 0}$, then the feedback controller in Eq. (32) guarantees that virus k is eradicated with at least an exponential rate of $\min_{t \in \mathbb{Z}_{\geq 0}} [1 - \sigma_1(\tilde{M}^k[t])]$.

Corollary 4. If we apply the mitigation strategy in Theorem 5 to all viruses $k \in [m]$, then the system in (2) converges to the set of healthy states at an exponential rate of $\max_{k \in [m]} \min_{t \in \mathbb{Z}_{\geq 0}} [1 - \sigma_1(\tilde{M}^k[t])]$.

Remark 6. The control strategy proposed in Theorem 5 can be interpreted as follows: if the healing rate with respect to virus k of each subpopulation is appropriately increased according to the susceptible level, for example by distributing effective medication, medical supplies, and healthcare workers to each subpopulation, then virus k will be eradicated at an exponential rate.

Our feedback controller (32) is an improvement with respect to similar *open-loop* control schemes for networked SIS models in (Gracy et al., 2020; Liu et al., 2019; Ye, Liu, Anderson, & Cao, 2021), since it does not require full information of the system states. More specifically, for the feedback gain design, because the recovered state can be calculated using $r_i[t] = h \sum_{q=0}^t \sum_{k=1}^m \gamma_i^k x_i^k[q]$, we only need knowledge of the susceptible level $s_i[t]$ or knowledge of the infected level, $x_i^k[t]$ for all $k \in [m]$, $i \in [n]$. Furthermore, compared to the control strategies proposed in (Gracy et al., 2020; Liu et al., 2019; Ye et al., 2021)

Table 1

For the network in Fig. 2, disease parameters corresponding to the Omicron variant.

β_{ij}^1	FR	IT	CH	AT	DE
FR	0.08	0.15	0.24	0	0.06
IT	0.15	0.12	0.13	0.11	0
CH	0.24	0.13	0.25	0.05	0.04
AT	0	0.09	0.05	0.11	0.15
DE	0.06	0	0.04	0.14	0.09
γ_i^1	0.15	0.23	0.17	0.25	0.2
$x^1[0]$	0.005	0.01	0.0075	0.0025	0.0075
c^1	0.4	0.4	0.4	0.4	0.4

Table 2

For the network in Fig. 2, disease parameters corresponding to the Delta variant.

β_{ij}^2	FR	IT	CH	AT	DE
FR	0.02	0.05	0.04	0	0.01
IT	0.05	0.06	0.07	0.02	0
CH	0.04	0.07	0.04	0.03	0.05
AT	0	0.03	0.04	0.09	0.07
DE	0.01	0	0.05	0.07	0.06
γ_i^2	0.095	0.12	0.1	0.15	0.13
$x^2[0]$	0.001	0.002	0.0035	0.002	0.001
c^2	0.3	0.3	0.3	0.3	0.3

whose boosted healing rates maintain constant values, our distributed feedback controller, since it updates the healing rates in response to the infection level in a subpopulation, is capable of allocating the medical resource more efficiently. In the next section, among other results, we explore the performance of our distributed feedback controller by relying only on the *estimates* of the system states in addition to the actual system states.

6. Simulations

In this section, we consider the special case of two competing variants of the SARS-CoV-2 virus, (i.e. $m = 2$): Delta and Omicron spreading over the network depicted in Fig. 2 (Zipfel, Colizza, & Bansal, 2021). We choose the two variants of the SARS-CoV-2 virus because they cause patients to display similar symptoms, such as fever, coughing, and headache (Antonelli et al., 2022). We consider a network of 5 nodes, where each node represents a country in Europe: France, Italy, Switzerland, Austria, and Germany; there exists an edge between two nodes if the countries that the nodes represent share a border with each other. The system parameters corresponding to the two variants of SARS-CoV-2 virus are listed in Tables 1 and 2, respectively. This section includes no real data; however, the viral spreading parameters are inspired by the behavior of the viruses (Shrestha, Foster, Rawlinson, Tedla, & Bull, 2022). That is, the model parameters are chosen so that the Omicron variant is more contagious than Delta. We also acknowledge that these two variants are not necessarily competitive; there are cases where people have been infected by both variants. However, the samples of co-infection of both variants are very uncommon (Bolze et al., 2022), thus can be ignored in the population for the time scales being considered in these simulations.

The evolution of the infection levels of both viruses are illustrated in Fig. 3. We plot the total fraction of individuals who exhibit similar symptoms that are caused due to either of the viruses, namely $y_i[t]$, with $i = [5]$, from the observation model in Eq. (4); see Fig. 4.

We then estimate the infection level by using the Luenberger observer in Eq. (23), with $\hat{s}[t]$ and $\hat{r}[t]$ calculated via Eqs. (21) and (22), respectively. The initial conditions are provided in Table 3. To investigate the impact of the observer gain L_i^k on the



Fig. 2. Contact graph for the spread of the Omicron and Delta variants of SARS-CoV-2 virus.

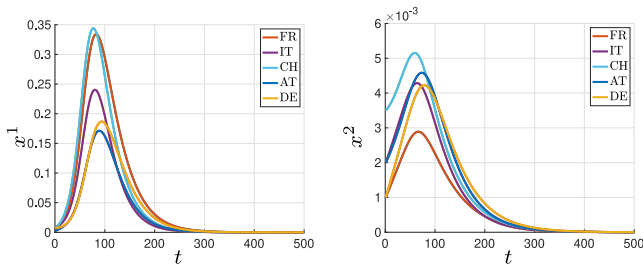


Fig. 3. Evolution of infection level of the Omicron variant in each country (left); Evolution of infection level of the Delta variant in each country (right).

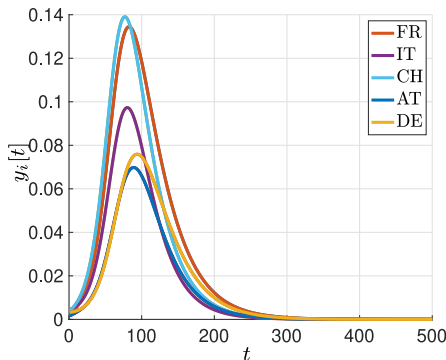


Fig. 4. Total proportion of individuals who show similar symptoms from both variants in each country.

estimation error, we simulate the Luenberger observer's performance with a scaled observer gain ηL_i^k , where $L_i^k = 1$, for all $i \in [5], k \in [2]$. Let t^* denote the time instant such that the aggregated estimation error $\frac{1}{mn} \sum_{k=1}^m \sum_{i=1}^n |x_i^k[t] - \hat{x}_i^k[t]| < 0.01$, for all $t > t^*$. In Fig. 5, we illustrate how the point in time at which the estimation error converges (t^*) depends on the gain of the observer (η) for this system. We can see that as the observer gain increases, the convergence time first decreases and then increases; eventually, t^* no longer exists because a large observer gain can cause the estimated infection level to exceed 1, that is, the estimated system states cease to be well-defined.

In Fig. 5, when the scale of the observer gain $\eta > 3.3$, the estimated infection level $\hat{x}_i^k[t] \rightarrow \infty$ as t increases.

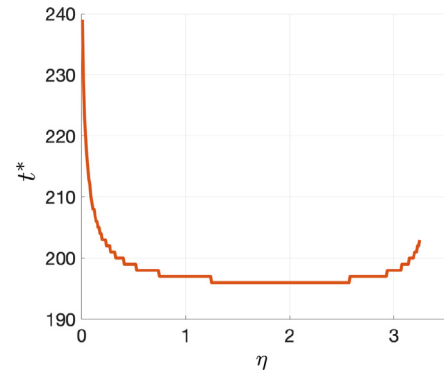


Fig. 5. The scale of the observer gain vs the time for the estimation error to converge to zero.

Table 3

Initial conditions assumed for the Luenberger observer (23).

	FR	IT	CH	AT	DE
$\hat{x}^1[0]$	0.0037	0.0075	0.0056	0.0019	0.0056
$\hat{x}^2[0]$	0.0005	0.001	0.002	0.001	0.0005

We next focus on finding the feasible observer gain L^k such that the systems states are well-defined. We set $\tau^1 = 0.1$ and $\tau^2 = 0.3$, where τ^k , for each $k \in [2]$, is as defined in Theorem 4. For different l^k values, we calculate the corresponding L^k values using LMI (31). It turns out that setting $l^k = 10^{10}$ for both viruses ensures that all the estimated system states are well-defined. Note that the feasible value of l^k is not unique. For our subsequent simulations, we set $l^k = 10^{10}$, and, by solving the LMI (31) in Corollary 3, we obtain the observer gains L^k , $k = 1, 2$,

$$L^1 = \begin{bmatrix} 0.101223398722677 \\ 0.0928658303375023 \\ 0.112524328507691 \\ 0.0860241631907317 \\ 0.0843783515296357 \end{bmatrix}, L^2 = \begin{bmatrix} 0.0853417070451051 \\ 0.0879525432030855 \\ 0.0898592088737154 \\ 0.0885900881539504 \\ 0.0897704274804431 \end{bmatrix}. \quad (34)$$

Notice that $|L^2| < |L^1|$. Thus, we need less compensation from the observer gain for the system state estimation of Delta than for that of Omicron.

We now explore whether we are able to compute the observer gains for all viruses through LMI (31), when the assumptions in Theorem 3 are not satisfied. If we set $\gamma_{DE}^2 = \gamma_{DE}^1 = 0.2$ in Table 2, then the LMI (31) becomes infeasible for the Delta variant, hence, we are not able to accurately estimate the infection level for the Delta variant. Therefore, when the healing rates of different viruses are identical at a node, we are unable to find the appropriate observer gain, which can be interpreted as saying that we are not able to estimate the system states of each virus if the viruses are indistinguishable (i.e., the recovery rates for the two viruses are the same) from each other at any subpopulation, consistent with Theorem 3.

To evaluate the performance of our proposed estimation algorithm, we simulate the state estimation errors, with observer gains from (34) and initial conditions from Table 3, in Fig. 6. We can see that, in Fig. 6 (top), the estimation error of the Omicron variant is negligible compared to its infection level, and converges to zero before its infection level does. In Fig. 6 (bottom), the estimation error of the Delta variant starts to die out around $t = 250$; while the infection levels converge to zero when $t > 300$, as seen in Fig. 3 (right). Hence, the simulation results in Fig. 6 are consistent with Theorem 4.

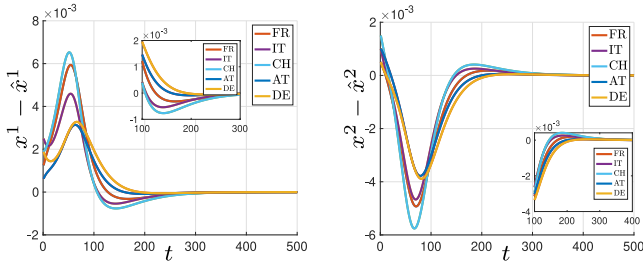


Fig. 6. Estimation error of infection level of the Omicron variant in each country (top); Estimation error of infection level of the Delta variant in each country (bottom).

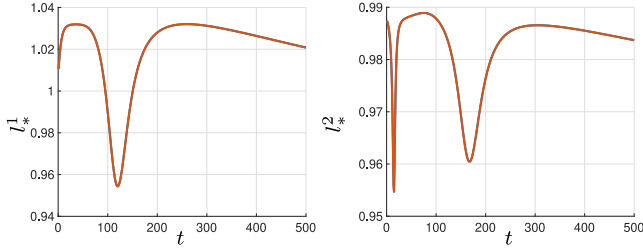


Fig. 7. Lipschitz-like constant l^k_* of the Omicron variant (left) and (right).

Next, we leverage the simulation results in Fig. 6 to evaluate the sufficient condition in Theorem 2. Solving the condition in Theorem 2 for s_i for France ($i = \text{FR}$) gives that $s_{\text{FR}} < \sim 0.134$. Recall that Theorem 2 only provides a sufficient condition and, therefore, as we can see by the estimation results, the condition is not very tight since the state estimation performs well before s_{FR} drops below 0.134.

Remark 7. Note that the sufficient condition in Theorem 2 is not necessary for strong local observability.

Inspired by (28), we now explore how to properly choose the Lipschitz-like constant l^k for our model by denoting:

$$l^k_*[t] = \frac{\|w^k[t]\|}{\|x^k[t] - \hat{x}^k[t]\|}, \quad (35)$$

for each $k \in [2]$. In Fig. 7, we plot the Lipschitz-like constant $l^k_*[t]$ of our network (see Fig. 2) for both viruses over time. Recall that in order to fulfill the condition in Assumption 6, we must choose l^k such that, for each $k \in [2]$:

$$l^k \geq \frac{\|w^k[t]\|}{\|x^k[t] - \hat{x}^k[t]\|}, \quad (36)$$

for all $t \in \mathbb{Z}_{\geq 0}$. Our choice of $l^k = 10^{10} > l^k_*[t]$ for all $t \in \mathbb{Z}_{\geq 0}$, as seen in Fig. 7, satisfies the requirement of Assumption 6.

In Fig. 8, we implement the distributed feedback mitigation strategy of Eq. (32) over all the countries in our network. We can see that, consistent with Theorem 5, our proposed mitigation strategy is able to eradicate the spread of both viruses at an exponential rate. Moreover, maintaining the mitigation strategy over the network throughout time ensures that no subsequent infection wave occurs. We then explore the state feedback controller of Eq. (32) which utilizes the estimated system states instead of the actual system states, i.e., the control input is calculated as:

$$\bar{u}_i^k[t] = \hat{s}_i[t] \sum_{j=1}^n \beta_{ij}^k, \quad (37)$$

where $\hat{s}_i[t]$ is obtained through the estimation algorithm in Eq. (23). In Fig. 9, we plot the infection level for both viruses using

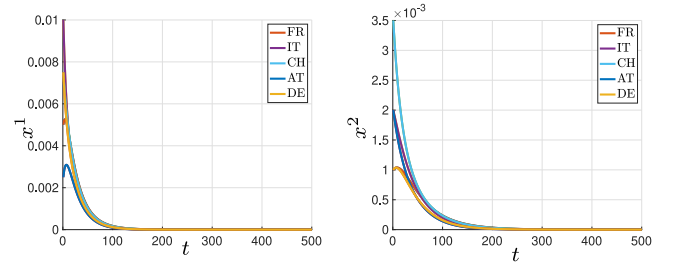


Fig. 8. Evolution of the infection level of the Omicron variant in each country with the distributed feedback control strategy in Eq. (32) (left); Evolution of the infection level of the Delta variant in each country the distributed feedback control strategy in Eq. (32) (right).

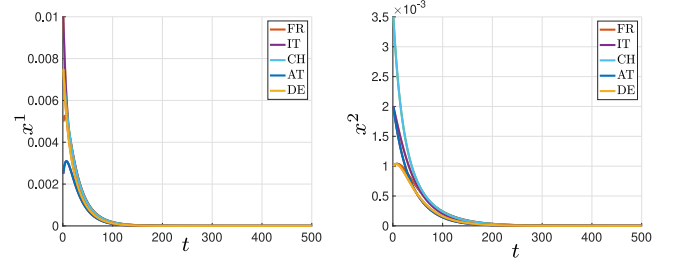


Fig. 9. Evolution of the infection level of the Omicron variant in each country with the distributed mitigation strategy with a control input from Eq. (37) (left); Evolution of the infection level of the Delta variant in each country with the distributed mitigation strategy with a control input from Eq. (37) (right).

the mitigation strategy (32) with the control input (37) applied. It can be seen that the feedback controller with the estimated states is able to achieve the same goal, namely, eradicating each virus at an exponential rate. Even though Theorem 5 did not prove that the feedback controller that uses the estimated susceptible level Eq. (37) eradicates the viruses at an exponential rate, it seems that our estimation algorithm in Eq. (23) is able to reconstruct the susceptible level accurately in order to enable rapid eradication in this case.

7. Conclusion

This paper has proposed a novel discrete-time networked multi-virus SIR model. We have provided two sufficient conditions for each virus to be eradicated exponentially fast. Then, we have specified a sufficient, but not necessary, condition for the two-virus system to be strongly locally observable at an arbitrary system state. We have subsequently presented a necessary and sufficient condition for the system to be strongly locally observable at a specific system state, namely when there are no susceptible individuals in the network. We have proposed a distributed Luenberger observer which estimates the system states. Thereafter, we have provided a method for choosing the observer gain such that the estimation error of each virus converges to zero asymptotically. We have introduced a distributed feedback mitigation strategy that ensures that each virus is eradicated at an exponential rate. In the simulations, we have utilized the Luenberger state observer proposed to estimate the system states, and the results illustrate that the Luenberger observer is suitable for state estimation of our model.

For future work, we plan to extend the strong local observability result for an arbitrary state to systems with more than two viruses. We also intend to find conditions which ensure that the rate of convergence of the observer is faster than that of the controller. We plan to explore the system state estimation problem in

the presence of measurement noise to emulate practical scenarios where there is error in the collected data. Further, we also hope to formulate and solve the parameter estimation problem for this model and then leverage these tools to validate the model on real epidemiological data. We also aim to study more complex models, such as the susceptible–exposed–infected–recovered and susceptible–asymptomatic–infected–recovered models, which incorporate additional compartments for exposed or asymptomatic individuals. Finally, we aim to extend the nonlinear distributed state estimator design to time-varying networks.

Appendix

The proofs of the technical results are provided here.

A.1. Proof of Lemma 1

Proof of statement (1). We assume that, for some arbitrary t , the following holds: $s_i[t], x_i^k[t], r_i[t] \in [0, 1]$, for all $k \in [m]$ and $s_i[t] + \sum_{k=1}^m x_i^k[t] + r_i[t] = 1$. By repeating the same steps from the Base Case except replacing 0 and 1 with k and $k+1$, we can write $s_i[t+1], x_i^k[t+1], r_i[t+1] \in [0, 1]$, for all $k \in [m]$ and $s_i[t+1] + \sum_{k=1}^m x_i^k[t+1] + r_i[t+1] = 1$. Therefore, by induction, we can prove that $s_i[t], x_i^k[t], r_i[t] \in [0, 1]$, for all $k \in [m]$ and $s_i[t] + \sum_{k=1}^m x_i^k[t] + r_i[t] = 1$ for all $i \in [n]$ and $t \in \mathbb{Z}_{\geq 0}$. By Assumption 1, the result holds for all $t \in \mathbb{Z}_{\geq 0}$.

Proof of statement (2). From (1) and Assumption 2 we know that $-h(s_i[t] \sum_{k=1}^m \sum_{j=1}^n \beta_{ij}^k x_j^k[t]) \leq 0$ for all $t \in \mathbb{Z}_{\geq 0}$. Thus, we have $s_i[t+1] \leq s_i[t]$ for all $i \in [n]$ and $t \in \mathbb{Z}_{\geq 0}$. ■

A.2. Proof of Corollary 1

From Lemma 3, (11), and (15), the rate of convergence of virus k is upper-bounded by $\sqrt{1 - \frac{\mu_3^k}{\mu_2^k}}$. We then need to show that the rate is well-defined, which is $\sqrt{1 - \frac{\mu_3^k}{\mu_2^k}} \in [0, 1)$. Since $\mu_2^k > 0$ and $\mu_3^k > 0$, it is sufficient to show that $\mu_2^k \geq \mu_3^k$.

Since P^k is positive definite and $(M^k)^\top P^k M^k$ is nonnegative definite, we have

$$\mu_3^k I \leq P^k - (M^k)^\top P^k M^k \leq P^k \leq \mu_2^k I, \quad (38)$$

from which $\mu_2^k \geq \mu_3^k$ and the rate of convergence $\sqrt{1 - \frac{\mu_3^k}{\mu_2^k}}$ is well-defined. ■

A.3. Proof of Proposition 1

Define $\epsilon^k[t] := 1 - \sigma_1(\tilde{M}^k[t])$. Hence, we obtain that $\epsilon^k[t] > 0$ for all $k \in [m]$ and $t \in \mathbb{Z}_{\geq 0}$. Since we have $x^k[t+1] = \tilde{M}^k[t]x^k[t]$ and $x^k[t] \geq 0$ for all $t \in \mathbb{Z}_{\geq 0}$, we can write

$$\frac{\|x^k[t+1]\|}{\|x^k[t]\|} = \frac{\|\tilde{M}^k[t]x^k[t]\|}{\|x^k[t]\|} \leq \sigma_1(\tilde{M}^k[t]), \quad (39)$$

which results in $\|x^k[t+1]\| \leq (1 - \epsilon^k[t])\|x^k[t]\|$ for all t . Recall that, $\epsilon^k[t] > 0$ for all $k \in [m]$ and $t \in \mathbb{Z}_{\geq 0}$. Then, we obtain that, for all $x_i^k[0] \in [0, 1]^n$,

$$\|x^k[t]\| \leq [1 - \min_{\forall t \in \mathbb{Z}_{\geq 0}} \epsilon^k[t]]^t \|x^k[0]\| \leq e^{-t \min_{\forall t \in \mathbb{Z}_{\geq 0}} \epsilon^k[t]} \|x^k[0]\|,$$

where the second inequality is due to Bernoulli's inequality (Carothers, 2000). Hence, $x^k[t]$ converges to $\mathbf{0}$ with at least an exponential rate of $\min_{\forall t \in \mathbb{Z}_{\geq 0}} \epsilon^k[t]$. ■

A.4. Proof of Theorem 2

When there are two competing viruses, namely $m = 2$, the observability matrix can be written as:

$$\mathcal{O}[t] = \begin{bmatrix} C^1 & C^2 \\ C^1 \tilde{M}^1 & C^2 \tilde{M}^2 \end{bmatrix}. \quad (40)$$

Note that by construction and Assumption 4, C^1 is a positive diagonal matrix and therefore is invertible and full rank. Consequently, we can study the rank of $\mathcal{O}[t]$ by examining

$$\begin{bmatrix} C^1 & C^2 \\ C^1 \tilde{M}^1 & C^2 \tilde{M}^2 \end{bmatrix} \begin{bmatrix} I & -(C^1)^{-1}C^2 \\ \mathbf{0} & I \end{bmatrix} \\ = \begin{bmatrix} C^1 & \mathbf{0}^{n \times n} \\ C^1 \tilde{M}^1 & C^2 \tilde{M}^2 - C^1 \tilde{M}^1 (C^1)^{-1}C^2 \end{bmatrix},$$

since multiplication by a full rank matrix preserves rank. Therefore, we need to show that $C^2 \tilde{M}^2 - C^1 \tilde{M}^1 (C^1)^{-1}C^2$ has full rank. This matrix simplifies to

$$h[C^2(S[t]B^2 + \Gamma^1 - \Gamma^2) - C^1 S[t]B^1 (C^1)^{-1}C^2] \quad (41)$$

and its i th row has the following structure:

$$h[s_i[t](c_i^2 \beta_{i1}^2 - \frac{c_i^1 c_1^2}{c_1^1} \beta_{i1}^1) \cdots \\ c_i^2 (\gamma_i^1 - \gamma_i^2 + s_i[t](\beta_{ii}^2 - \beta_{ii}^1)) \cdots \\ s_i[t](c_i^2 \beta_{in}^2 - \frac{c_i^1 c_n^2}{c_n^1} \beta_{in}^1)].$$

Since, by Assumption 2, $h > 0$, and by assumption,

$$c_i^2 |\gamma_i^1 - \gamma_i^2 + s_i[t](\beta_{ii}^2 - \beta_{ii}^1)| > s_i[t] \sum_{j \neq i} |c_i^2 \beta_{ij}^2 - \frac{c_i^1 c_j^2}{c_j^1} \beta_{ij}^1|,$$

for all $i \in [n]$, we have, from the Gershgorin circle theorem, that (41) has no zero eigenvalues and, therefore, is full rank. Consequently, the mapping in Eq. (18) is injective. Thus, by Lemma 5, the system in Eq. (4), for $m = 2$, is strongly locally observable at $s[t]$. ■

A.5. Proof of Theorem 3

Proof. Note that the sufficiency result, for $m = 2$, follows directly from Theorem 2.

For $m > 2$, we have the following proof. If $s[t] = \mathbf{0}$, then the observability matrix $\mathcal{O}[t]$ defined in Eq. (18) becomes

$$\mathcal{O}_0[t] = [\mathcal{O}_0^1[t] \quad \mathcal{O}_0^2[t] \quad \cdots \quad \mathcal{O}_0^m[t]], \quad (42)$$

where

$$\mathcal{O}_0^k[t] = \begin{bmatrix} C^k \\ C^k(I - h\Gamma^k) \\ C^k(I - h\Gamma^k)^2 \\ \vdots \\ C^k(I - h\Gamma^k)^{m-1} \end{bmatrix}, \quad (43)$$

for all $k \in [m]$. From the assumptions $h \sum_{k=1}^m \gamma_i^k \leq 1$, $h > 0$, and $\gamma_i^k > 0$ for all $i \in [n]$, $k \in [m]$, we obtain that $1 - h\gamma_i^k \in (0, 1)$ for all $i \in [n]$, $k \in [m]$. In addition, by Assumption 4, $c_i^k \in (0, 1]$ for all $i \in [n]$, $k \in [m]$, we can conclude that the entries of Eq. (42): $c_i^k(1 - h\gamma_i^k) \in (0, 1)$ for all $i \in [n]$, $k \in [m]$.

We let $\mathbf{0}^n := [0 \ 0 \ \cdots \ 0]^{1 \times n}$ and $\mathbf{0}^0 := \emptyset$. Consider Eq. (43) and recall that every block matrix is diagonal; hence, Eq. (42) is the concatenation of a set of block diagonal matrices.

For all $i \in [n]$, the i th row of the observability matrix (42) can be written as:

$$[\mathbf{0}^{i-1} \quad c_i^1 \quad \mathbf{0}^{n-i} \quad \mathbf{0}^{i-1} \quad c_i^2 \quad \mathbf{0}^{n-i} \quad \dots \quad \mathbf{0}^{i-1} \quad c_i^m \quad \mathbf{0}^{n-i}]$$

which is linearly independent with the $(i + ln)$ -th row of (42) for all $l \in [m - 1]$:

$$[\mathbf{0}^{i-1} \quad c_i^1(1 - h\gamma_i^1)^l \quad \mathbf{0}^{n-i} \quad \dots \quad \mathbf{0}^{i-1} \quad c_i^m(1 - h\gamma_i^m)^l \quad \mathbf{0}^{n-i}]$$

under our assumption that, for each $i \in [n]$, $\gamma_i^k \neq \gamma_i^{k_0}$ for all $k, k_0 \in [m]$ and $k \neq k_0$. Thus, the observability matrix in Eq. (42) has full row rank. Since the observability matrix is a square matrix, we conclude that the observability matrix, $\mathcal{O}_0[t]$, is full rank. Therefore, the mapping in Eq. (18) when $s[t] = \mathbf{0}$ is injective. Note that mn is the dimension of $\mathbf{X}[t]$. Hence, by Lemma 5, the competing virus model in (4) is strongly locally observable at $s[t] = \mathbf{0}$.

For necessity, assume that there exist $i \in [n]$ such that for some $k, k_0 \in [m]$, with $k \neq k_0$, $\gamma_i^k = \gamma_i^{k_0} =: \gamma_i$. Consequently, the $(i + (k - 1)n)$ -th column of (42) becomes

$$[\mathbf{0}^{i-1} \quad c_i^k \quad \mathbf{0}^{n-i} \quad \mathbf{0}^{i-1} \quad c_i^k(1 - h\gamma_i) \quad \mathbf{0}^{n-i} \quad \dots \quad \mathbf{0}^{i-1} \quad c_i^k(1 - h\gamma_i)^{m-1} \quad \mathbf{0}^{n-i}]^\top,$$

which is linearly dependent on the $(i + (k_0 - 1)n)$ -th column of (42):

$$[\mathbf{0}^{i-1} \quad c_i^{k_0} \quad \mathbf{0}^{n-i} \quad \mathbf{0}^{i-1} \quad c_i^{k_0}(1 - h\gamma_i) \quad \mathbf{0}^{n-i} \quad \dots \quad \mathbf{0}^{i-1} \quad c_i^{k_0}(1 - h\gamma_i)^{m-1} \quad \mathbf{0}^{n-i}]^\top.$$

Therefore, the observability matrix, $\mathcal{O}_0[t]$, is not full rank. Thus, the mapping in Eq. (18), when $s[t] = \mathbf{0}$, is not injective. Hence, by Lemma 5, the competing virus model in (4) is not strongly locally observable at $s[t] = \mathbf{0}$. ■

A.6. Proof of Theorem 4

From Assumption 6, we can write that

$$(w^k[t])^\top w^k[t] \leq (l^k)^2 (e^k[t])^\top e^k[t]$$

which can be rewritten as

$$[(e^k[t])^\top \quad (w^k[t])^\top] \Phi^k \begin{bmatrix} e^k[t] \\ w^k[t] \end{bmatrix} \leq 0, \quad (44)$$

where

$$\Phi^k = \begin{bmatrix} -(l^k)^2 I & \mathbf{0} \\ \mathbf{0} & I \end{bmatrix}.$$

By utilizing the Schur complement (Zhang, 2006), and defining $A^k = (M^k - L^k C^k)^\top Q^k (M^k - L^k C^k)$, where Q^k is a symmetric positive-definite matrix by assumption, we can reorganize Eqs. (29) and (30) as

$$\begin{bmatrix} A^k - Q^k + \tau^k (l^k)^2 I & (M^k - L^k C^k)^\top Q^k \\ Q^k (M^k - L^k C^k) & Q^k - \tau^k I \end{bmatrix} \prec 0, \quad (45)$$

which yields the following:

$$\Omega^k - \tau^k \Phi^k \prec 0, \quad (46)$$

where

$$\Omega^k = \begin{bmatrix} A^k - Q^k & (M^k - L^k C^k)^\top Q^k \\ Q^k (M^k - L^k C^k) & Q^k \end{bmatrix}.$$

We now consider the candidate Lyapunov function $V_2^k(e^k, [t]) = (e^k[t])^\top Q^k(e^k[t])$, where Q^k is the positive-definite matrix in (45). We have that

$$V_2^k(e^k, [t]) > 0, \text{ for all } e^k[t] \neq \mathbf{0}.$$

For $e^k \neq \mathbf{0}$, we have

$$\begin{aligned} \Delta V_2^k &= V_2^k(e^k[t+1], t) - V_2^k(e^k[t], t) \\ &= \left[(M^k - L^k C^k) e^k[t] + w^k[t] \right]^\top Q^k \\ &\quad \left[(M^k - L^k C^k) e^k[t] + w^k[t] \right] - (e^k[t])^\top Q^k e^k[t] \\ &= (e^k[t])^\top A^k e^k[t] \\ &\quad + (w^k[t])^\top Q^k (M^k - L^k C^k) e^k[t] \\ &\quad + (e^k[t])^\top (M^k - L^k C^k)^\top Q^k w^k[t] + (w^k[t])^\top Q^k w^k[t] \\ &\quad - (e^k[t])^\top Q^k e^k[t] \\ &= (e^k[t])^\top (A^k - Q^k) e^k[t] \\ &\quad + (w^k[t])^\top Q^k (M^k - L^k C^k) e^k[t] \\ &\quad + (e^k[t])^\top (M^k - L^k C^k)^\top Q^k w^k[t] \\ &\quad + (w^k[t])^\top Q^k w^k[t] \\ &= \begin{bmatrix} (e^k[t])^\top & (w^k[t])^\top \end{bmatrix} \Omega^k \begin{bmatrix} e^k[t] \\ w^k[t] \end{bmatrix} \\ &< \begin{bmatrix} (e^k[t])^\top & (w^k[t])^\top \end{bmatrix} \tau^k \Phi^k \begin{bmatrix} e^k[t] \\ w^k[t] \end{bmatrix} \\ &\leq 0, \end{aligned} \quad (47)$$

where inequality (47) follows from (46), and inequality (48) follows from inequality (44). Therefore, by Lyapunov's direct method (Vidyasagar, 2002), the estimation error of the Luenberger observer (23) for virus k is globally asymptotically stable. ■

A.7. Proof of Theorem 5

By substituting Eqs. (32) and (33) into (2b), we obtain

$$x_i^k[t+1] = x_i^k[t] + h \left\{ s_i[t] \sum_{j=1}^n \beta_{ij}^k x_j^k[t] - \left[s_i[t] \sum_{j=1}^n \beta_{ij}^k + \gamma_i^k \right] x_i^k[t] \right\}. \quad (49)$$

By the assumption that $h\tilde{\gamma}_i^k[t] < 1, \forall i \in [n], t \in \mathbb{Z}_{\geq 0}$, the infection states will never go negative, by an analogous argument to the one made in the proof of Lemma 1. The state transition matrix of (49) can be written as

$$\tilde{M}^k[t] = I + h \left[S[t] B^k - (S[t] \text{diag}(B^k \mathbf{1}^{n \times 1}) + \text{diag}(\gamma_i^k)) \right].$$

The entries of the i th row of $\tilde{M}^k[t]$, therefore, are

$$\tilde{m}_{ii}^k[t] = 1 + h \left[s_i[t] \beta_{ii}^k - s_i[t] \sum_{j=1}^n \beta_{ij}^k - \gamma_i^k \right],$$

$$\tilde{m}_{ij}^k[t] = h s_i[t] \beta_{ij}^k,$$

Therefore,

$$\|\tilde{M}^k[t]\|_\infty = \max_{i \in [n]} \tilde{m}_{ii}^k[t] + \sum_{j \neq i} \tilde{m}_{ij}^k[t] = \max_{i \in [n]} 1 - h\gamma_i^k < 1, \quad (50)$$

where the inequality holds by Assumption 3. Further,

$$\begin{aligned} \|\tilde{M}^k[t]\|_1 &= \max_{i \in [n]} \tilde{m}_{ii}^k[t] + \sum_{j \neq i} \tilde{m}_{ji}^k[t] \\ &= \max_{i \in [n]} 1 + h \left[\sum_{j \neq i} s_j[t] \beta_{ji}^k - s_i[t] \sum_{j \neq i} \beta_{ij}^k - \gamma_i^k \right] \end{aligned}$$

$$\leq \max_{i \in [n]} 1 + h \left[\sum_{j \neq i}^n s_j[t] \beta_{ji}^k - \gamma_i^k \right] \quad (51)$$

$$< 1, \quad (52)$$

where (51) holds by the nonnegativity of $s_i[t]$ and (52) by assumption.

Recall that the following relationship holds (Horn & Johnson, 2012, page 365)

$$(\sigma_1(A)) = \|A\|_2 \leq (\|A\|_1 \|A\|_\infty)^{\frac{1}{2}} \leq \max(\|A\|_1, \|A\|_\infty). \quad (53)$$

Thus, by (50) and (52),

$$\sigma_1(\tilde{M}^k[t]) < 1,$$

for all $t \in \mathbb{Z}_{\geq 0}$. Therefore, by Proposition 1, virus k converges to zero with an exponential rate of $\min_{t \in \mathbb{Z}_{\geq 0}} [1 - \sigma_1(\tilde{M}^k[t])]$. ■

References

- Allen, L. J., & van den Driessche, P. (2008). The basic reproduction number in some discrete-time epidemic models. *Journal of Difference Equations and Applications*, 14(10–11), 1127–1147.
- Antonelli, M., Pujol, J. C., Spector, T. D., Ourselin, S., & Steves, C. J. (2022). Risk of long COVID associated with Delta versus Omicron variants of SARS-CoV-2. *The Lancet*, 399(10343), 2263–2264.
- Arcak, M., & Kokotović, P. (2001). Nonlinear observers: A circle criterion design and robustness analysis. *Automatica*, 37(12), 1923–1930.
- Atkinson, K. E. (2008). *An introduction to numerical analysis*. John Wiley & Sons.
- Bara, G. I., Zemouche, A., & Boutayeb, M. (2005). Observer synthesis for Lipschitz discrete-time systems. In *Proc. IEEE international symposium on circuits and systems* (pp. 3195–3198).
- Belongia, E. A., & Osterholm, M. T. (2020). COVID-19 and flu, a perfect storm. *Science*, 368(6496), 1163–1163.
- Benedictow, O. J., & Benedictow, O. L. (2004). *The black death, 1346–1353: The complete history*. Boydell & Brewer.
- Berge, T., Lubuma, J.-S., Moremedi, G., Morris, N., & Kondera-Shava, R. (2017). A simple mathematical model for Ebola in Africa. *Journal of Biological Dynamics*, 11(1), 42–74.
- Bolze, A., Basler, T., White, S., Dei Rossi, A., Wyman, D., Dai, H., Roychoudhury, P., Greninger, A. L., Hayashibara, K., Beatty, M., et al. (2022). Evidence for SARS-CoV-2 Delta and Omicron co-infections and recombination. *Med*, 3(12), 848–859.
- Brauer, F., Castillo-Chavez, C., & Feng, Z. (2019). *Mathematical models in epidemiology: Vol. 32*, Springer.
- Brauer, F., Feng, Z., & Castillo-Chavez, C. (2009). Discrete epidemic models. *Mathematical Biosciences & Engineering*, 7(1), 1–15.
- Calafiore, G. C., Novara, C., & Possieri, C. (2020). A modified SIR model for the COVID-19 contagion in Italy. In *Proc. conference on decision and control* (pp. 3889–3894).
- Carothers, N. L. (2000). *Real analysis*. Cambridge University Press.
- Chang, H.-J. (2017). Estimation of basic reproduction number of the Middle East respiratory syndrome coronavirus (MERS-CoV) during the outbreak in South Korea, 2015. *Biomedical Engineering Online*, 16(1), 1–11.
- Chen, Y.-C., Lu, P.-E., Chang, C.-S., & Liu, T.-H. (2020). A time-dependent SIR model for COVID-19 with undetectable infected persons. *IEEE Transactions on Network Science and Engineering*, 7(4), 3279–3294.
- Cheng, K., & Leung, P. (2007). What happened in China during the 1918 influenza pandemic? *International Journal of Infectious Diseases*, 11(4), 360–364.
- Coburn, B. J., Wagner, B. G., & Blower, S. (2009). Modeling influenza epidemics and pandemics: Insights into the future of swine flu (H1N1). *BMC Medicine*, 7(1), 1–8.
- Colizza, V., Barrat, A., Barthélemy, M., & Vespignani, A. (2007). Predictability and epidemic pathways in global outbreaks of infectious diseases: The SARS case study. *BMC Medicine*, 5(1), 1–13.
- Gracy, S., Anderson, B. D., Ye, M., & Uribe, C. A. (2024). Competitive networked bivirus SIS spread over hypergraphs. In *Proc. American control conference* (pp. 4409–4415).
- Gracy, S., Paré, P. E., Liu, J., Sandberg, H., Beck, C. L., Johansson, K. H., & Başar, T. (2025). Modeling and analysis of a coupled SIS bi-virus model. *Automatica*, 171, Article 111937.
- Gracy, S., Paré, P. E., Sandberg, H., & Johansson, K. H. (2020). Analysis and distributed control of periodic epidemic processes. *IEEE Transactions on Control of Network Systems*, 8(1), 123–134.
- Horn, R. A., & Johnson, C. R. (2012). *Matrix analysis*. Cambridge University Press.
- Ito, H. (2022). Strict smooth Lyapunov functions and vaccination control of the SIR model certified by ISS. *IEEE Transactions on Automatic Control*, 67(9), 4514–4528.
- Johnson, N. P., & Mueller, J. (2002). Updating the accounts: Global mortality of the 1918–1920 Spanish influenza pandemic. *Bulletin of the History of Medicine*, 105–115.
- Liu, J., Paré, P. E., Nedić, A., Tang, C. Y., Beck, C. L., & Başar, T. (2019). Analysis and control of a continuous-time bi-virus model. *IEEE Transactions on Automatic Control*, 64(12), 4891–4906.
- Ma, S., Zhang, F., Zhou, F., Li, H., Ge, W., Gan, R., Nie, H., Li, B., Wang, Y., Wu, M., et al. (2021). Metagenomic analysis reveals oropharyngeal microbiota alterations in patients with COVID-19. *Signal Transduction and Targeted Therapy*, 6(1), 1–11.
- Mei, W., Mohagheghi, S., Zampieri, S., & Bullo, F. (2017). On the dynamics of deterministic epidemic propagation over networks. *Annual Reviews in Control*, 44, 116–128.
- Meyerowitz-Katz, G., & Merone, L. (2020). A systematic review and meta-analysis of published research data on COVID-19 infection fatality rates. *International Journal of Infectious Diseases*, 101, 138–148.
- Mitra, A., & Sundaram, S. (2018). Distributed observers for LTI systems. *IEEE Transactions on Automatic Control*, 63(11), 3689–3704.
- Niazi, M. U. B., & Johansson, K. H. (2022). Observer design for the state estimation of epidemic processes. In *Proc. conference on decision and control* (pp. 4325–4332).
- Nijmeijer, H. (1982). Observability of autonomous discrete time non-linear systems: a geometric approach. *International Journal of Control*, 36(5), 867–874.
- Nowak, M. (1991). The evolution of viruses. competition between horizontal and vertical transmission of mobile genes. *Journal of Theoretical Biology*, 150(3), 339–347.
- Osthus, D., Hickmann, K. S., Caragea, P. C., Higdon, D., & Del Valle, S. Y. (2017). Forecasting seasonal influenza with a state-space SIR model. *The Annals of Applied Statistics*, 11(1), 202.
- Panovska-Griffiths, J., Kerr, C. C., Stuart, R. M., Mistry, D., Klein, D. J., Viner, R. M., & Bonell, C. (2020). Determining the optimal strategy for reopening schools, the impact of test and trace interventions, and the risk of occurrence of a second COVID-19 epidemic wave in the UK: A modelling study. *The Lancet Child & Adolescent Health*, 4(11), 817–827.
- Paré, P. E., Liu, J., Beck, C. L., Nedić, A., & Başar, T. (2017). Multi-competitive viruses over static and time-varying networks. In *Proc. American control conference* (pp. 1685–1690).
- Paré, P. E., Vrabac, D., Sandberg, H., & Johansson, K. H. (2020). Analysis, online estimation, and validation of a competing virus model. In *Proc. American control conference* (pp. 2556–2561).
- Pepin, K. M., Lambeth, K., & Hanley, K. A. (2008). Asymmetric competitive suppression between strains of dengue virus. *BMC Microbiology*, 8(1), 1–10.
- Perks, R., & Schneck, N. (2021). COVID-19 in artisanal and small-scale mining communities: Preliminary results from a global rapid data collection exercise. *Environmental Science & Policy*, 121, 37–41.
- Poland, A. M., Vennema, H., Foley, J. E., & Pedersen, N. C. (1996). Two related strains of feline infectious peritonitis virus isolated from immunocompromised cats infected with a feline enteric coronavirus. *Journal of Clinical Microbiology*, 34(12), 3180–3184.
- Prakash, B. A., Beutel, A., Rosenfeld, R., & Faloutsos, C. (2012). Winner takes all: Competing viruses or ideas on fair-play networks. In *Proc. 21st international conference on world wide web* (pp. 1037–1046).
- Rader, B., Gertz, A., Iuliano, A. D., Gilmer, M., Wronski, L., Astley, C. M., Sewalk, K., Varrelman, T. J., Cohen, J., Parikh, R., et al. (2022). Use of at-home COVID-19 tests—United States, August 23 2021–March 12 2022. *Morbidity and Mortality Weekly Report*, 71(13), 489.
- Rantzer, A. (2011). Distributed control of positive systems. In *Proc. conference on decision and control and European control conference* (pp. 6608–6611).
- Rugh, W. J. (1996). *Linear system theory: Vol. 2*, NJ: Prentice Hall Upper Saddle River.
- Russell, T. W., Hellewell, J., Jarvis, C. I., Van Zandvoort, K., Abbott, S., Ratnayake, R., Flasche, S., Eggo, R. M., Edmunds, W. J., Kucharski, A. J., et al. (2020). Estimating the infection and case fatality ratio for coronavirus disease (COVID-19) using age-adjusted data from the outbreak on the Diamond Princess cruise ship, February 2020. *Eurosurveillance*, 25(12), Article 2000256.
- Sahneh, F. D., & Scoglio, C. (2014). Competitive epidemic spreading over arbitrary multilayer networks. *Physical Review E*, 89(6), Article 062817.

- Santos, A., Moura, J. M., & Xavier, J. M. (2015). Bi-virus SIS epidemics over networks: Qualitative analysis. *IEEE Transactions on Network Science and Engineering*, 2(1), 17–29.
- She, B., Leung, H. C., Sundaram, S., & Paré, P. E. (2021). Peak infection time for a networked SIR epidemic with opinion dynamics. In *Proc. conference on decision and control* (pp. 2104–2109).
- She, B., Sundaram, S., & Paré, P. E. (2022). A learning-based model predictive control framework for real-time SIR epidemic mitigation. In *Proc. American control conference* (pp. 2565–2570).
- Shrestha, L. B., Foster, C., Rawlinson, W., Tedla, N., & Bull, R. A. (2022). Evolution of the SARS-CoV-2 Omicron variants BA. 1 to BA. 5: Implications for immune escape and transmission. *Reviews in Medical Virology*, 32(5), Article e2381.
- Smith, K. D., & Bullo, F. (2022). Convex optimization of the basic reproduction number. *IEEE Transactions on Automatic Control*, 68(7), 4398–4404.
- Snow, J. (1855). *On the mode of communication of cholera*. John Churchill.
- Sontag, E. D. (1979). On the observability of polynomial systems, I: Finite-time problems. *SIAM Journal on Control and Optimization*, 17(1), 139–151.
- Tennis, P. F., Moe, C. L., Liu, P., E. Miller, S., Lindesmith, L., Baric, R. S., Le Pendu, J., & Calderon, R. L. (2008). Norwalk virus: how infectious is it? *Journal of Medical Virology*, 80(8), 1468–1476.
- Van den Driessche, P. (2008). Spatial structure: Patch models. *Mathematical Epidemiology*, 179–189.
- Van Mieghem, P., Omic, J., & Kooij, R. (2009). Virus spread in networks. *IEEE/ACM Transactions on Networking*, 17(1), 1–14.
- Vidyasagar, M. (2002). *Nonlinear systems analysis*. SIAM.
- World Health Organization (2020). Global coronavirus (2019-nCoV). <https://www.who.int/emergencies/diseases/novel-coronavirus-2019>. (Accessed 08 December 2022).
- World Health Organization (2022). 2009 H1N1 pandemic. <https://www.cdc.gov/flu/pandemic-resources/2009-h1n1-pandemic.html>. (Accessed 08 December 2022).
- Ye, M., Liu, J., Anderson, B. D., & Cao, M. (2021). Applications of the Poincaré–Hopf theorem: Epidemic models and Lotka–Volterra systems. *IEEE Transactions on Automatic Control*, 67(4), 1609–1624.
- Zhang, F. (2006). *The Schur complement and its applications: Vol. 4*, Springer Science & Business Media.
- Zhang, C., Gracy, S., Başar, T., & Paré, P. E. (2022). A networked competitive multi-virus SIR model: Analysis and observability. *IFAC-PapersOnLine*, 55(13), 13–18.
- Zhang, C., Leung, H., Butler, B. A., & Paré, P. E. (2024). Estimation and distributed eradication of SIR epidemics on networks. *IEEE Transactions on Control of Network Systems*, 11(2), 756–768.
- Zino, L., Ye, M., & Anderson, B. D. (2024). Modeling and analyzing competitive epidemic diseases with partial and waning virus-specific and cross-immunity. *IFAC Journal of Systems and Control*, 28, Article 100262.
- Zipfel, C. M., Colizza, V., & Bansal, S. (2021). The missing season: The impacts of the COVID-19 pandemic on influenza. *Vaccine*, 39(28), 3645–3648.



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